01 / Right-angle sidecar: packing and reference fixture planes

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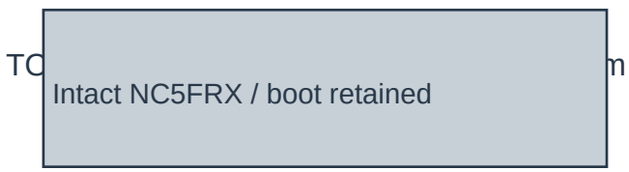
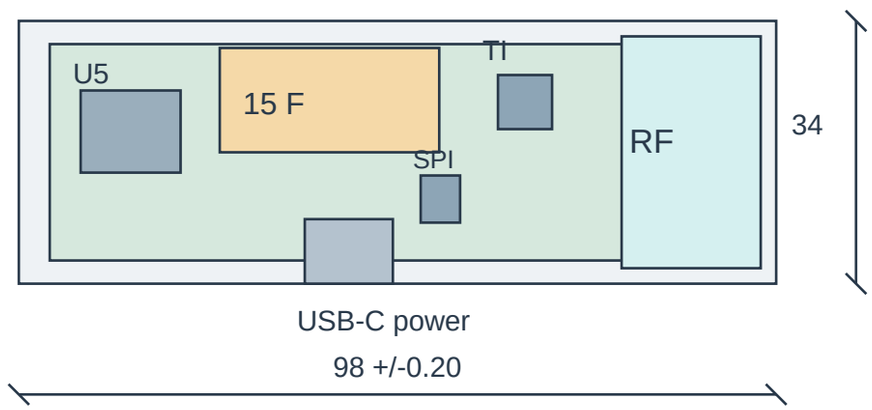
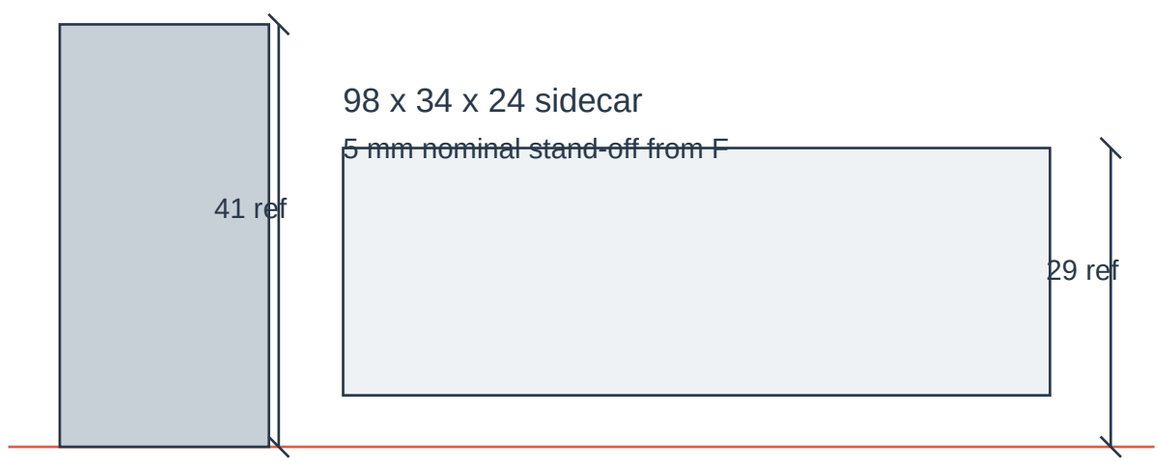




SIDE / fixture reference plane F = z0



F: actual recess / insertion / connector pitch UNKNOWN

PACKING / x-y coordinates and full critical stack in canonical design

U5 max: 12.93 x 10.63 x 3.55; TI RGZ nominal: 7 x 7; SPI NOR DFN max: 5.10 x 6.10 x 0.60.

Cap max: diameter 13.5 x 28.4; PCB gap1.0 / lid gap1.5 nominal; cradle detailed design open.

USB: 11.4 x 8.35 x 3.66 envelope; footprint width reserve 12.3; shell aperture 14 x 6.

Cavity minimum width 29.5 minus PCB maximum 28.1 = 1.4 total lateral clearance.

Latch: retain 18 mm approach envelope; test finger action with power and DMX cables fitted.

Antenna keepout x78..96: no screws, capacitor or ground copper. External metal gap requires RF validation.

98 mm retained: 25 x 6 power reservation does NOT prove complete EDA placement; routing/cradle remain open.

Supplier components shown as envelopes. Verify actual footprints, latch approach, harness and fixture CAD before tooling.

See canonical design for tolerances, assembly requirements, release gates and primary-source links.

02 / Rotary USB barrel: fixed core, lock and independent stops

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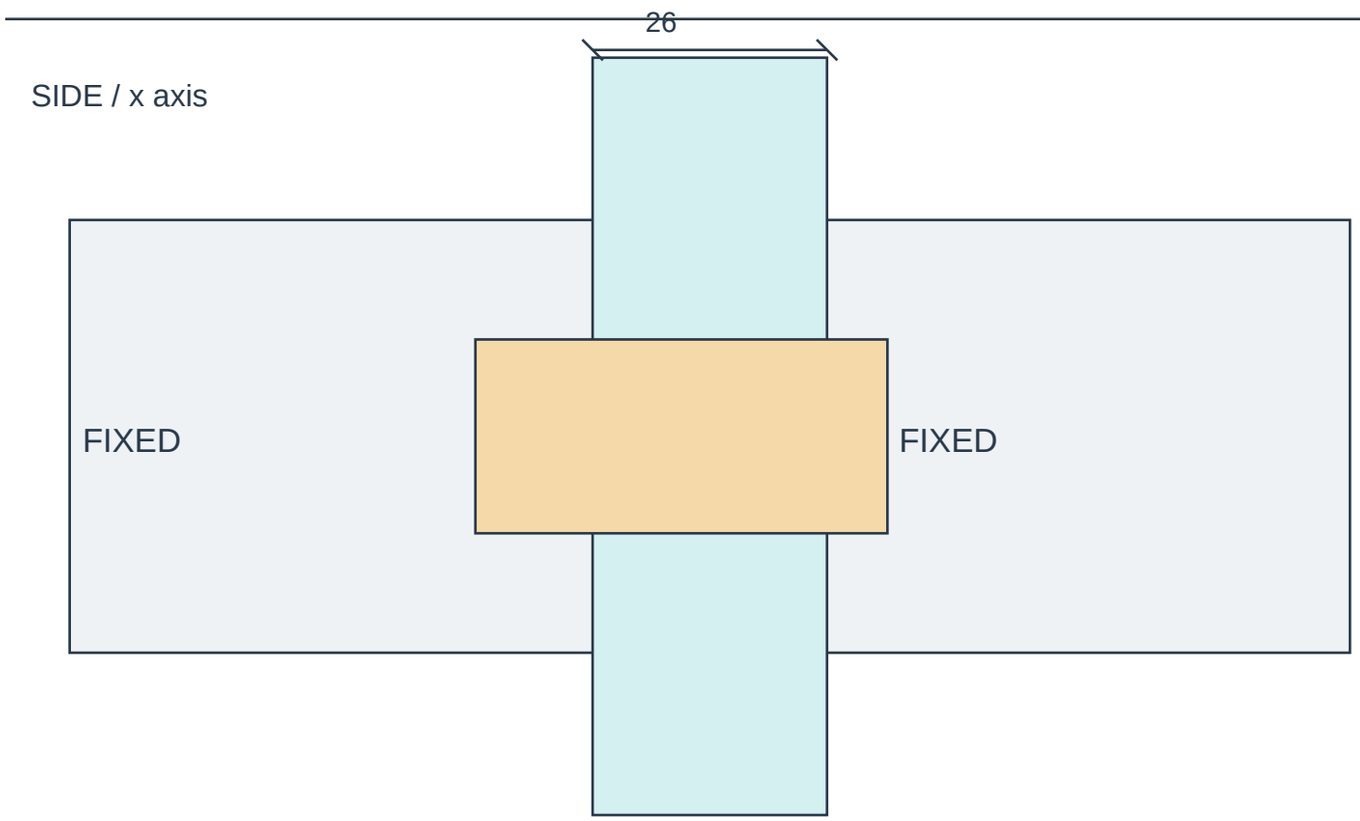

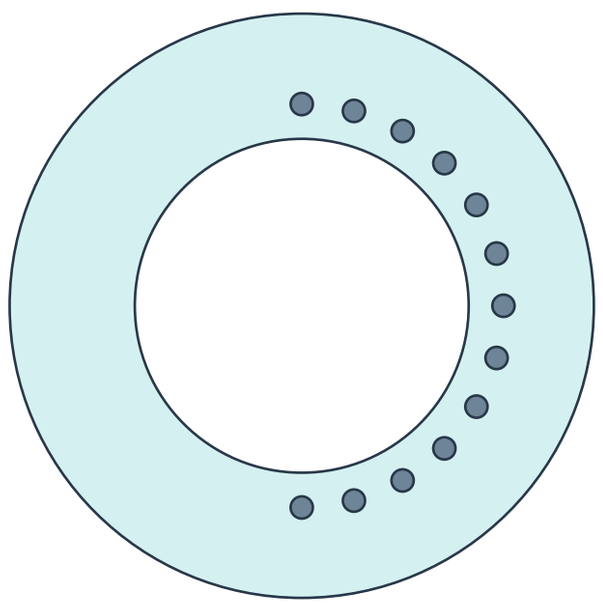

Core OD48 / collar OD84 / normal USB-C output 5 V, 1.5 A

L/N/PE and PSU do not rotate. Source / full assembly approval pending.

13 index holes / 15 degree spacing

R29; diameter 3.20 +0.05/0

Pin diameter 3.00 -0.02/0

Pull 4.2 mm to release; spring return

Independent R30 stop slot: width4.30; rounded R2.15 ends; diameter4 peg through diameter4.10 cheek bore.

Nominal stop travel180.57 degrees; acceptance180 +/-1. Lock release cannot bypass stop.

Lock guide B19: OD3.80 / ID3.20 x9; x58..67; divider/covers seal to sleeve; attachment still open.

USB window x65.5..78; wire-lane edge x64.8; board and receptacle lowered1 mm to clear guide.

Target: lock1 N m, stop2 N m, axial100 N, cable40 N. Retainers/endcaps/slider and qualification open.

Supplier components shown as envelopes. Verify actual footprints, latch approach, harness and fixture CAD before tooling.

See canonical design for tolerances, assembly requirements, release gates and primary-source links.

03 / Secondary wire cassette: rolling-loop calculation and exploded order

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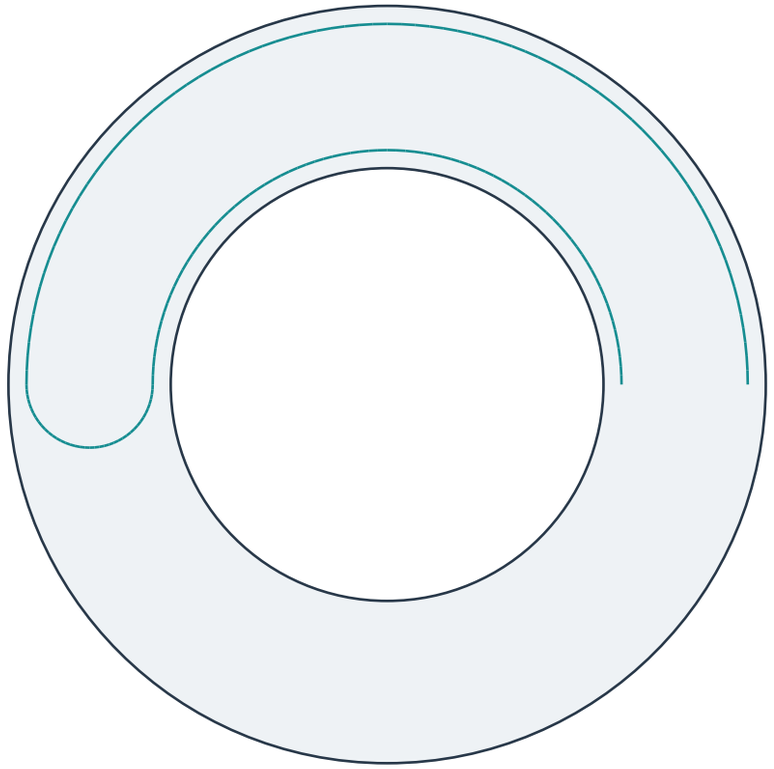


Neutral: two arcs + R7 reversing fold

Two independent axial wire lanes, no mains in cassette.

ri=26, ro=40; R=(40-26)/2=7 mm.

theta=-90..+90 degrees; phi=pi+40 theta/66.

Loop angle phi=125.45..234.55 degrees.

Lactive=pi(26+40+7)=229.34 mm.

Add 10 mm clamped tail each end:249.34 mm.

24 AWG high-strand wire OD <=1.2 mm.

Supplier dynamic bend qualification <=7 mm required.

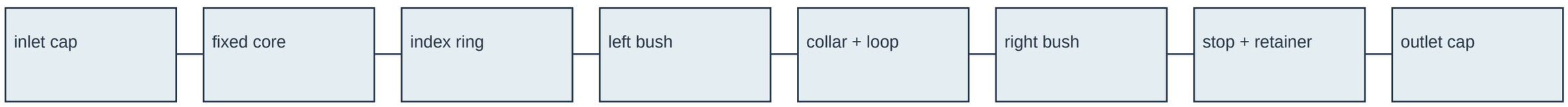
No twist in ideal routing; motion test still mandatory.

Slots1.6 / divider0.8; seal divider/covers at B19 sleeve.

B19-wire gap0.50 nominal /0.375 allocated worst case.

Termination/controller losses must be added.

EXPLODED ASSEMBLY ORDER / corresponding STEP is spatially exploded



Lock ring / pin remain independent of the mains path. Assemble at keyed neutral; verify both stops before closing.

Supplier components shown as envelopes. Verify actual footprints, latch approach, harness and fixture CAD before tooling.

See canonical design for tolerances, assembly requirements, release gates and primary-source links.